

RioClear Initiative: Integrated Surface Water Contamination and Drinking Water Compliance Surveillance Along the U.S.–Mexico Border

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Introduction

Across the 13 Texas border counties along the Rio Grande there is over 2.87 million residents that rely on public water systems and adjacent surface-water resources. Our project objective is to synthesize regulatory data from the U.S. EPA and water sample data from the USGS/TCEQ into a unified, open-source dashboard to locate failing drinking-water systems, identify surface-water contamination hotspots, and compute an objective priority score to guide intervention.

Overview

RioClear Initiative combines two public datasets on one map to answer one question: where are drinking-water systems failing, where is surface water contaminated, and where should action start first?

It covers the 13 Texas counties that border the Rio Grande, upstream to downstream: El Paso, Hudspeth, Presidio, Brewster, Terrell, Val Verde, Kinney, Maverick, Webb, Zapata, Starr, Hidalgo and Cameron.

The project is three Python files. Two scripts download and clean the data. One Streamlit app turns it into an interactive dashboard:

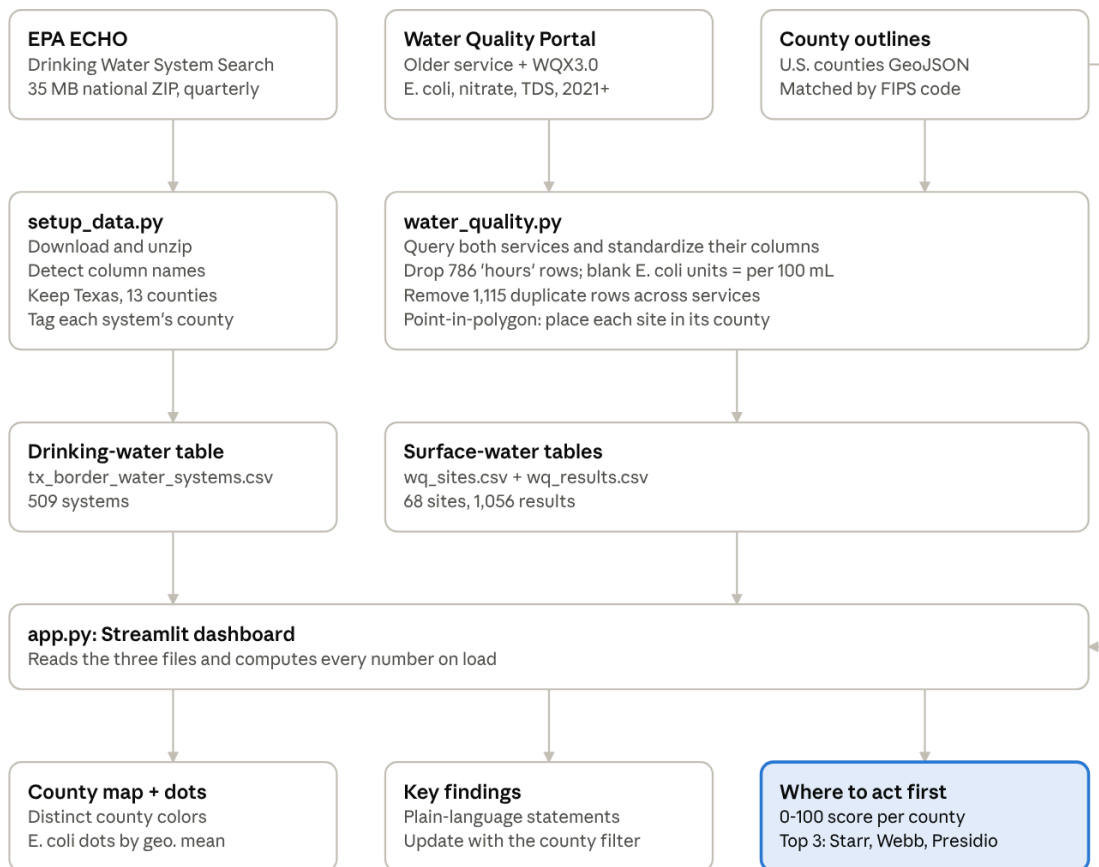
File	Job	Output
setup_data.py	Downloads EPA drinking-water compliance data and keeps the 13 border counties	tx_border_water_systems.csv (509 systems)
water_quality.py	Downloads surface-water samples, merges two services, places each site in a county	wq_sites.csv (68 sites), wq_results.csv (1,056 results)
app.py	Builds the map, findings, priority ranking and tables	The live dashboard

Every number on screen comes from a transparent data pipeline and a simple scoring formula which can be traced back to a public record.

Live dashboard: rioclear-initiative.streamlit.app · Code: github.com/by-tayo/rioclear-initiative

How it works: the pipeline

Two public datasets become one county priority ranking



The data Pipeline has 3 sources, 2 scripts, and 1 dashboard

Read top to bottom: two scripts clean the public data into three tables, and the dashboard computes every number, including the priority ranking, from those tables.

Data sources

Both datasets are public, free and published by federal agencies; we did not create or estimate any measurements.

Source	Run by	What we used	Coverage
EPA ECHO Drinking Water System Search Dataset	U.S. EPA, from the Safe Drinking Water Information System (SDWIS)	One row per public water system: name, counties served, population served, health-based violation	Nationwide, refreshed quarterly; filtered to 509 systems in the 13 counties

Source	Run by	What we used	Coverage
		flag, serious-violator status	
Water Quality Portal	USGS and EPA, with state agencies such as the Texas Commission on Environmental Quality (TCEQ)	Sample results for E. coli, nitrate and total dissolved solids, plus each site's coordinates	January 2021 to June 2026; 68 sites in the 13 counties

The Water Quality Portal runs two services. Its older one [stopped receiving new USGS data in March 2024](#); newer USGS data goes to the WQX3.0 service. We query both and merge them, which is why our data runs into 2026.

County outlines come from a public U.S. county map file (GeoJSON) published by Plotly, matched to each county by its FIPS code.

Part 1: drinking-water data (setup_data.py)

This script turns a 35 MB national file into a 509-row table of border-county water systems.

1. **Download.** It fetches the ECHO Drinking Water System Search ZIP. If the file is already on disk, it skips the download.
2. **Unzip and discover.** It lists the CSV files inside and reads only their header rows.
3. **Detect columns.** Instead of hard-coding column names, it looks for the best match: an exact preferred name first (such as `StateCode` or `CountiesServed`), then any column containing `STATE` or `COUNT`. This keeps the script working if EPA renames a column.
4. **Filter to Texas.** It keeps rows whose state value is `TX`, `TEXAS` or `48` (the Texas FIPS code).
5. **Filter to the border.** It keeps systems whose counties-served text names one of the 13 counties, using whole-word matching so a name inside another word does not count.
6. **Tag a county.** Each system gets a `border_county` column: the first border county it serves. Systems serving several counties are counted once, under that first county.
7. **Save.** It writes `tx_border_water_systems.csv` and `available_columns.txt`, a list of every field for reference.

Result: 509 systems, from 142 in El Paso down to 3 in Terrell.

Part 2: surface-water data (water_quality.py)

This script turns 3,244 raw rows from two services into 1,056 clean results at 68 sites.

1. **Ask both services.** One request each for sites (older service), results (older service) and results (WQX3.0, profile `basicPhysChem`). Every request names all 13 counties by FIPS code, the four measurement names, and a start date of January 1, 2021. Downloads are saved, so a re-run is instant; `--refresh` forces a new download.

2. **Standardize columns.** The two services name the same field differently (`MonitoringLocationIdentifier` vs `Location_Identifier`). The script maps both to one set: site, measurement, value, unit, date, organization, coordinates.
3. **Diagnose.** It prints raw row counts per measurement and service, and the E. coli units it saw. This is how we found the E. coli bug (next section).
4. **Clean.** It keeps numeric values only and labels each measurement. "Inorganic nitrogen (nitrate and nitrite)" is grouped under Nitrate.
5. **Fix E. coli units.** Rows in "hours" (786 rows, lab timing records) are dropped. Rows per 100 mL, CFU or MPN are kept. Rows with a blank unit (1,218) are kept and treated as per 100 mL, the standard way E. coli is reported.
6. **Remove duplicates.** The same sample often appears in both services; 1,115 duplicate rows (same site, measurement, date and value) were removed.
7. **Place each site in a county.** A point-in-polygon test checks which county outline contains the site's coordinates. 21 sites fell outside all 13 counties (mostly across the river or county lines) and were dropped.

The E. coli bug, and how we found it

Our first run returned zero E. coli results. The diagnostics showed about 1,000 E. coli rows per service, but most had a blank unit, and our first filter only kept units mentioning "100". Relaxing the rule for blank units recovered 450 E. coli results.

The dashboard (`app.py`)

The app reads the three CSV files and draws everything from them each time it loads, so new data means new results with no code changes.

Part	What it shows	How it works
Headline numbers	Systems, population served, systems with a health-based flag, serious violators	Sums over the selected counties
County map	The 13 counties, each in its own color, with its name at the center	Folium (Leaflet) map; label placed at the county's area-weighted centroid; auto-zooms to fit the selected counties
Hover box	County name, water systems, population served, serious violators	Per-county totals attached to each outline
Sampling-site dots	E. coli, nitrate or TDS per site	Circle markers on their own layer above the labels; click for a popup

Part	What it shows	How it works
Key findings	Plain-language statements with numbers	Computed from the data, so they change with the county filter
Where to act first	A 0-100 priority score per county, top 3 cards, full ranking	Formula in the next section
Systems table	Every system in the selected counties	Sorted by health flag, then violation quarters, then population

How the sampling dots are colored

- **E. coli:** by the site's geometric mean since 2021. Green is at or below 126 per 100 mL (EPA and Texas criterion for swimming and wading). Yellow is 126 to 399. Orange is above 399 (the Texas single-sample criterion).
- **Nitrate and TDS:** by which third of all sites the site's median falls in (lowest, middle, highest). These are relative ranks, not safety thresholds.

The geometric mean is used for E. coli because bacteria counts swing widely from one sample to the next; it is also the form the 126 criterion is written in.

Sidebar controls: county filter, map coloring (distinct colors or a heat scale by any metric), show or hide sampling sites, measurement picker, and map height for screenshots.

Where to act first: the priority score

Each county's score is the average of up to three signals, each rescaled so the lowest county scores 0 and the highest scores 1, then multiplied by 100.

Signal	Raw measure	Source
Health-flagged systems	Share of the county's systems with a health-based violation flag	EPA ECHO
Serious violators	Serious violators per 10 systems	EPA ECHO
E. coli above 126	Share of the county's E. coli sites with a geometric mean above 126 per 100 mL	Water Quality Portal

Rescaling (min-max) puts the three signals on the same 0-1 footing so no single one dominates:

Scaled value = (county value - lowest county) ÷ (highest county - lowest county)
Score = 100 × average of the scaled values

A county with no E. coli monitoring is averaged over the two drinking-water signals only. The "main driver" shown on each card is the signal with the highest scaled value.

Worked example: Starr, score 76

- Health-flagged systems: 4 of 22 = 18.2%, the highest of the 13 counties, so it scales to 1.00.
- E. coli above 126: 2 of 2 sites = 100%, the top of the range, so it scales to 1.00.
- Serious violators: 2 per 22 systems = 0.9 per 10, which scales to about 0.28 against the other counties.
- Score: $(1.00 + 0.28 + 1.00) / 3 \times 100 \approx 76$.

What the score does and does not capture

It counts how many sites are high, not how high they are. Webb's worst site (geometric mean 8,172) is about 65 times the 126 level, but only 3 of its 5 sites are high and few of its drinking-water systems are flagged, so Webb scores 39. We kept the formula simple on purpose and printed it on the dashboard so anyone can check it.

Key findings

Across the 13 counties, 23 drinking-water systems serving about 86,231 people carry health-based violation flags, and 24 of 57 surface-water sites exceed the E. coli level for safe swimming and wading.

Drinking water (EPA ECHO)

- 509 public water systems serve 2,872,711 people.
- 23 systems carry a health-based violation flag; they serve about 86,231 people.
- 14 systems are listed as serious violators.
- Starr has the highest share of flagged systems: 18.2% (4 of 22).

Surface water (Water Quality Portal, 2021 to 2026)

- 24 of 57 E. coli sites have a geometric mean above 126 per 100 mL.
- Rio Grande Below Laredo (Webb, TCEQ): geometric mean 8,172 over 9 samples, all above 399, latest June 26, 2024.
- Los Olmos Creek at SH 755 (Starr, TCEQ): geometric mean 2,151 over 4 samples, 75% above 399.
- Webb has the highest typical E. coli levels (median of its sites' geometric means).
- Hudspeth and Kinney have no E. coli monitoring in the data since 2021.

Top priority counties

County	Score	Main driver	Systems with health flag	Serious violators	People served by flagged systems	E. coli sites above 126	Worst site (geometric mean)
Starr	76	Health-flagged systems	4 of 22	2	18,918	2 of 2	2,151
Webb	39	E. coli above 126	2 of 40	4	871	3 of 5	8,172

County	Score	Main driver	Systems with health flag	Serious violators	People served by flagged systems	E. coli sites above 126	Worst site (geometric mean)
Presidio	38	Health-flagged systems	3 of 28	2	252	1 of 3	139

In Webb, the flagged systems serve only 871 of the county's 275,200 people. The finding there is about the river, not Laredo's tap water.

Limitations

This is a screening tool that shows where to look first; it is not an official water-quality assessment.

Limitation	Why it matters	How we handle it
Small samples	Starr's "2 of 2" is two sites; some sites have 4 to 9 samples	Findings are stated only for counties with monitoring
Mixed site types	Sites include the Rio Grande, creeks, canals and drainage ditches	We call them surface-water sites, not river sites; drains show where contamination enters the river
Blank E. coli units	1,218 rows had no unit and were treated as per 100 mL	The values checked out as normal bacteria counts (826, 2,151, 8,172)
Single-sample vs geometric-mean criteria	The 126 criterion applies to a geometric mean; 399 applies to single samples	We color by geometric mean and report the share of samples above 399 separately
Some sites are dated	One Cameron site's latest sample is from 2022	Each popup shows the latest sample date
Multi-county systems	A system serving two counties is counted under the first border county	Totals per county are close, not exact
Score counts, not severity	Webb's extreme 8,172 does not lift its score above Starr	The formula is printed on the dashboard; severity shows in the dots and popups

What the data does not say:

- It does not say anyone got sick from this water; we have no illness data.
- It does not say Laredo's tap water is unsafe; the high readings are in the river.
- It does not show wrongdoing by any person; it shows documented violations.

Next steps, what we learned, glossary

Next steps

1. Add county-level reportable illness data from Texas DSHS to test whether high-contamination counties show more waterborne illness.
2. Add trends over time: is each site getting better or worse since 2021?
3. Add data from the Mexican side of the river, where much of the shared water originates.
4. Label each site as river, tributary or drain, with a river-only filter.
5. Recommend new E. coli monitoring sites for Hudspeth and Kinney.

What we learned

- **Public APIs:** querying the Water Quality Portal with county codes, measurement names and dates, and handling two services with different column names.
- **Geospatial analysis:** GeoJSON county outlines, FIPS codes, point-in-polygon tests, and area-weighted centroids for labels.
- **Messy-data cleaning:** column detection, unit mismatches, and removing 1,115 duplicate rows.
- **Debugging with diagnostics:** printing raw counts by service and unit is what revealed the hidden E. coli data.
- **Dashboards:** Streamlit layouts, Folium maps, caching, and designing for screenshots.

Glossary

Term	Meaning
ECHO	EPA's Enforcement and Compliance History Online, which publishes compliance records
SDWIS	Safe Drinking Water Information System, EPA's database of public water systems
Health-based violation	A violation of a standard set to protect health, such as exceeding a contaminant limit
Serious violator	EPA's label for a system with serious or repeated unresolved violations
E. coli	Bacteria from human or animal waste; a marker of sewage contamination
CFU / MPN per 100 mL	Two lab methods for counting bacteria in 100 milliliters of water
Geometric mean	An average of multiplied values, suited to counts that vary by orders of magnitude
TDS	Total dissolved solids, a measure of salinity

Term	Meaning
FIPS code	A standard numeric code for each U.S. state and county (Texas = 48, Webb = 48479)
Point-in-polygon	A test of whether a map point lies inside a shape, such as a county outline
WQX3.0	The newer Water Quality Portal data format, which holds USGS data after March 2024

Sources

- [ECHO Data Downloads, U.S. EPA](#)
- [Water Quality Portal Web Services Guide](#)
- [dataRetrieval Status, USGS](#)